When Models Surprise Us

Emergent Abilities, Chain-of-Thought, and the Thinking Revolution

AI in Industry

2026

The Unifying Thread

"The dynamics of learning in LLMs violate our deepest intuitions."

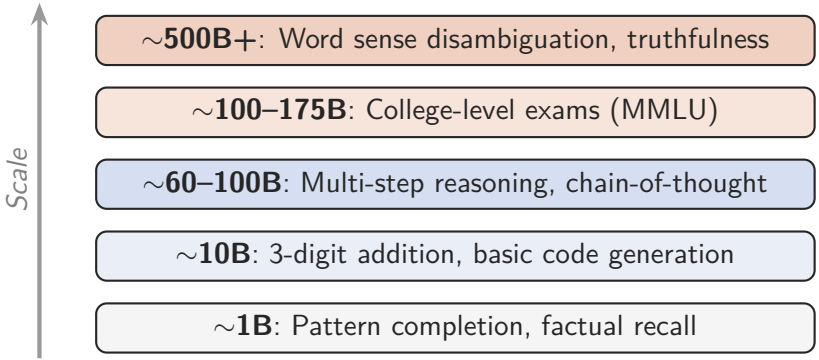
A. Emergent Abilities

Scale a model 10× and it suddenly writes poetry.

B. Chain-of-Thought & Thinking Models

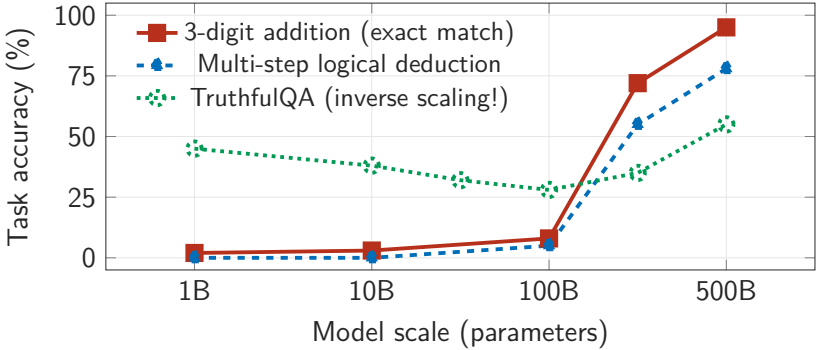
Ask a model to "think step by step" and it solves competition maths.

What Models Can Do at Each Scale



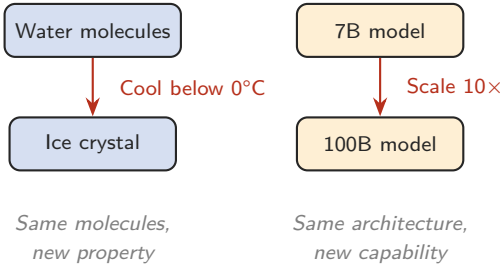
Approximate thresholds from *Wei et al., 2022, TMLR, Emergent Abilities of Large Language Models*.

Concrete Examples: From Nothing to Capability



3-digit addition: 0% → 80%+ in a narrow range. **TruthfulQA**: medium models are less truthful than small ones.

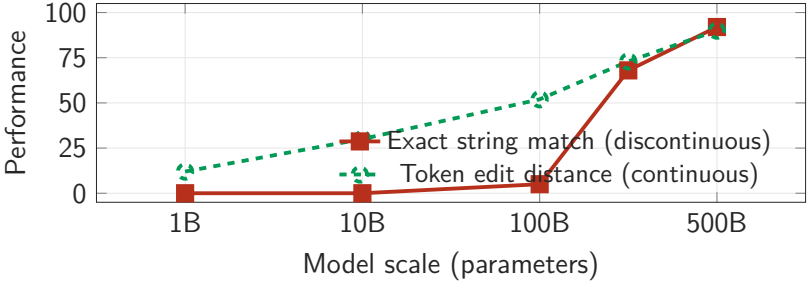
“More Is Different”



“The ability to reduce everything to simple fundamental laws does not imply the ability to start from those laws and reconstruct the universe.”

P.W. Anderson, 1972

Or Is It a Mirage?



92% of claimed emergent abilities in BIG-Bench appeared only under **discontinuous metrics**. Switch to a continuous metric ⇒ smooth improvement.

Schaeffer et al., 2023, NeurIPS Outstanding Paper, Are Emergent Abilities of LLMs a Mirage?

Capabilities can appear suddenly at scale.

But what if you could unlock reasoning *without* scaling the model?

Chain-of-Thought Prompting

Standard prompting:

```
Q: Roger has 5 tennis balls.
He buys 2 cans of 3 balls.
How many does he have?

A: 11
```

Chain-of-thought:

```
A: Roger started with 5.
2 cans × 3 = 6 new balls.
5 + 6 = 11.  The answer is 11.
```

Method	GSM8K
Standard	∼18%
Chain-of-Thought	∼58%
Prior SOTA	55%

Zero fine-tuning. Just 8 hand-written exemplars with reasoning steps.

Wei et al., 2022, NeurIPS, Chain-of-Thought Prompting Elicits Reasoning in LLMs

From One Trick to a Toolkit

Zero-shot CoT (Kojima et al., 2022)

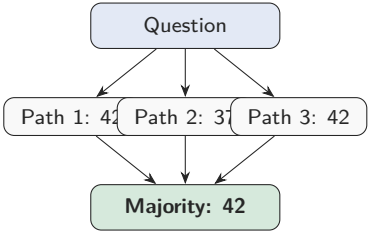
Just append:

"Let's think step by step."

GSM8K: 10.4% → **40.7%**
MultiArith: 17.7% → **78.7%**

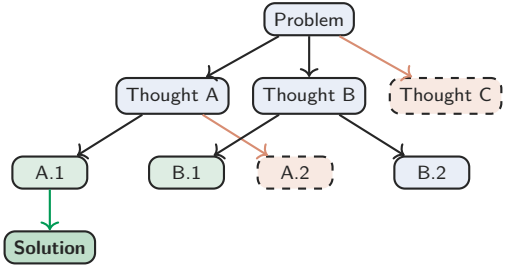
A single sentence, no exemplars, produces most of the benefit.

Self-Consistency (Wang et al., 2023)



Sample N paths, majority vote.
GSM8K: **+17.9%** over CoT.

Tree of Thoughts: Reasoning as Search



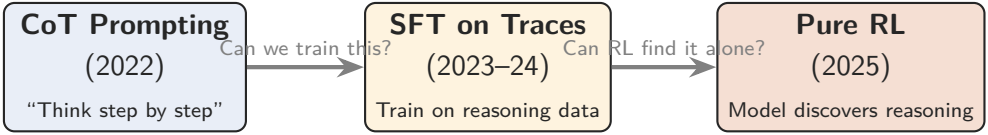
Generate multiple candidate thoughts at each step. The LLM evaluates each and uses BFS or DFS to explore, backtracking when stuck.

Game of 24:

CoT: **4%** ToT: **74%**

Yao et al., 2023, NeurIPS, Tree of Thoughts

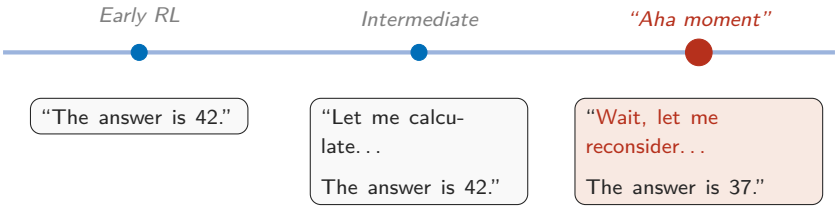
The Leap: From Prompting to Thinking Models



Wei et al., Kojima et al. *OpenAI o1* *DeepSeek-R1-Zero*

A prompting trick (2022) → trained reasoning models (2024) → models that discover reasoning from scratch via RL (2025). Under 3 years.

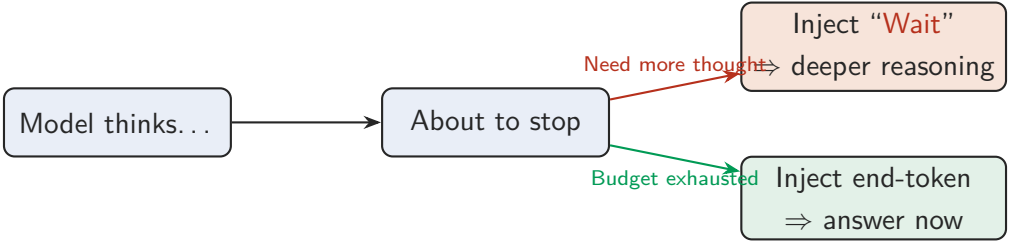
DeepSeek-R1-Zero: The "Aha Moment"



Pure RL, no human demos. The model discovered that pausing to self-correct improves answers. AIME 2024: 15.6% → **71.0%** (RL alone). Majority voting: **86.7%**, matching OpenAI o1.

DeepSeek-AI, 2025, Nature, DeepSeek-R1 Incentivizes Reasoning via RL

Budget Forcing: Controlling How Long a Model Thinks



Fine-tuned on only **1,000 examples**.

Training: **26 minutes** on 16 H100s.

Muennighoff et al., 2025, arXiv, s1: Simple Test-Time Scaling

Model	AIME
o1-preview	44.6%
s1-32B	50.0%
s1 + Wait×6	57.0%

How Do You Verify Reasoning?

Outcome Reward Models (ORMs)

Score the **final answer** only.
Binary: correct or incorrect.
Simple but coarse.

Process Reward Models (PRMs)

Score **each reasoning step**.
Catches errors early. 78% on MATH.

Lightman et al., 2024, ICLR, Let's Verify Step by Step

Generative PRMs

Use a long-CoT model to *verify* each step via its own reasoning. Strong verification, far fewer labels.

Why it matters

Process supervision outperforms outcome supervision. Scoring each step is key to scaling test-time compute.

The Test-Time Compute Landscape

Search-based

- Best-of-N
- Tree of Thoughts
- Self-Consistency
- Beam search

Refinement-based

- Self-correction
- Budget forcing
- Sequential revision
- "Wait" injection

Verification-based

- Outcome RMs (ORMs)
- Process RMs (PRMs)
- Generative PRMs
- Step-level scoring

Hybrid / Learned: o1, o3, DeepSeek-R1, MCTS+PRM

A small model with optimal test-time compute can outperform a 14× larger model (Snell et al., ICLR 2025 Oral). The optimal strategy is prompt-dependent.

Two Axes of Scaling

The capability frontier now has **two dimensions**:

Train-time compute

One-time capital expense.

Bigger models: GPT-2 $\rightarrow$ GPT-4.

More parameters, data, FLOPs.

Test-time compute

Per-query operating expense.

Think longer: o1, DeepSeek-R1.

Same model, more reasoning tokens.

Strategic implications:

- **High-volume, low-latency**: invest in a fast, large model.
- **High-stakes, low-volume**: smaller model + extended thinking.
- Test-time compute is most effective for medium-difficulty problems.

Questions to Take Home

1. If we cannot predict what the next model will be capable of, how should organisations plan for AI adoption?

2. Chain-of-thought evolved from a prompting trick to a training paradigm in under 3 years. What reasoning capabilities are next?

3. Test-time compute trades latency for accuracy. Which of your business decisions warrant a model that "thinks longer"?

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